

Package ‘dtab’

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Title Descriptive and Crosstabulation Tables for Survey Data

Version 0.1.0

Description Computes weighted and unweighted descriptive statistics (means, medians, sums, percentages, counts) broken down by one or more predictors, with optional confidence intervals, significance tests, and pairwise (letter-based) significance testing against survey-weighted designs via the 'survey' package. Reshapes the results into publication-ready crosstabulation, summary, battery-grid, and numeric-summary tables (via 'huxtable'), and exports them to Excel.

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tibble,
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openxlsx,
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utils

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calc_stats	<i>Compute weighted or unweighted descriptive statistics, broken down by zero or more predictors</i>
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Description

The engine every `pivot_*()`/`make_*()` function in this package builds on: one long-format tibble, one row per (outcome, category, predictor level, statistic) combination, computed either on a plain data frame (unweighted) or a survey design object (weighted statistics and design-based SEs/CIs available). Handles categorical, multicoded, and numeric outcomes; flat, nested, or multiple side-by-side predictor sets; optional SE/CI, p-values, and letter-based pairwise significance testing.

Usage

```
calc_stats(
  data,
  outcomes,
  predictors = NULL,
  statistics = c("count", "mean"),
  conf = NULL,
  base = NULL,
  pval = NULL,
  multicode = TRUE,
  pairwise = FALSE
)
```

Arguments

data	A data frame, or a survey design object (e.g. from <code>srvyr::as_survey_design()</code>) for weighted statistics.
outcomes	Character vector of outcome variable names.
predictors	NULL for no breakdown (Total only), a character vector/single variable name, a list of variable names for multiple side-by-side flat predictor sets, or <code>list(c(<outer>, <inner>))</code> for a nested predictor set. All predictors must be factors.
statistics	Character vector of statistic codes to compute, e.g. "perc", "count", "mean", "median", "sum", "min", "max", "range", "iqr", "sd", and their weighted w_-prefixed equivalents (weighted versions require a survey design as data).
conf	NULL, "se", or "ci" - adds a standard error or confidence interval alongside each estimate.
base	Optional base/filter expression restricting which cases each outcome is computed over.

pval	Optional p-value method for testing association between an outcome and a predictor (e.g. chi-square for categorical, ANOVA/ Shapiro-Wilk-gated for numeric).
multicode	If TRUE (default), an outcome coded as multiple binary/dummy columns sharing a common stem is treated as a single multicoded (select-all-that-apply) variable rather than several independent ones.
pairwise	If TRUE, runs add_pairwise_sig() internally and adds sig_letter/sig_diff columns (only for predictor levels, and only for statistics in pairwise_eligible_stats).

Value

A long-format tibble with one row per (outcome, category, predictor level, statistic): outcome/o_lab (variable name/label), o_cat (category, for categorical/multicoded outcomes), cross_break/p_cat1/p_lab (predictor variable/level/label, and p_cat2/p_lab2/predictor2 when a nested predictor set was used), stat (the statistic code), estimate (and estimate_se/ estimate_ci_low/estimate_ci_high if conf was requested), base (and base_description), plus p_value/sig_letter/sig_diff when requested. Feed this straight into [pivot_summary\(\)](#), [pivot_crosstab\(\)](#), [pivot_nested_crosstab\(\)](#), [pivot_battery\(\)](#), or [add_pairwise_sig\(\)](#) - or use one of the make_*() wrappers to skip calling calc_stats() directly.

See Also

[add_pairwise_sig\(\)](#), [pivot_summary\(\)](#), [pivot_crosstab\(\)](#), [pivot_nested_crosstab\(\)](#), [pivot_battery\(\)](#), [make_table\(\)](#)

export_tables_to_excel

Export any number of huxtables to one .xlsx file, one sheet per table

Description

Writes each table to its own sheet, and - for 2 or more tables only - adds an auto-generated, hyper-linked Contents sheet placed first. For a single table, use [export_table_to_excel\(\)](#) instead (a thin wrapper around this function that skips the Contents sheet entirely).

Usage

```
export_tables_to_excel(
  hux_list,
  file,
  sheet_names = NULL,
  titles = NULL,
  stem_cols = NULL,
  max_stem_width = 45
)
```

Arguments

<code>hux_list</code>	A list of huxtable objects, as returned by any of this package's <code>format_*</code> () functions (or <code>make_*</code> () with <code>formatted = TRUE</code>).
<code>file</code>	Output path for the .xlsx file.
<code>sheet_names</code>	Optional character vector of sheet names, one per table. Defaults to "Table 1", "Table 2", ...
<code>titles</code>	Optional character vector of title-row text, one per table (also used for the Contents sheet's listing, if shown).
<code>stem_cols</code>	Optional column width override(s) for the stub column(s), passed through to the underlying worksheet writer.
<code>max_stem_width</code>	Maximum stub column width, in Excel column-width units. Default 45.

Value

Invisibly, the `openxlsx` workbook object. Primarily called for its side effect of writing file.

See Also

[export_table_to_excel\(\)](#)

`export_table_to_excel` *Export a single huxtable to a one-sheet .xlsx file*

Description

Convenience wrapper around [export_tables_to_excel\(\)](#) for the common single-table case - delegates to it with a length-1 list, which (via that function's own guard) produces a workbook with exactly the one named sheet, no Contents tab alongside it.

Usage

```
export_table_to_excel(ht, file, sheet = "Table 1", title = NULL)
```

Arguments

<code>ht</code>	A single huxtable object, as returned by any of this package's <code>format_*</code> () functions (or <code>make_*</code> () with <code>formatted = TRUE</code>).
<code>file</code>	Output path for the .xlsx file.
<code>sheet</code>	Sheet name. Default "Table 1".
<code>title</code>	Optional title-row text.

Value

Invisibly, the `openxlsx` workbook object. Primarily called for its side effect of writing file.

See Also

[export_tables_to_excel\(\)](#)

make_battery_table	<i>Build a battery/grid table in one call</i>
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Description

Single entry point chaining [calc_stats\(\)](#) straight to [pivot_battery\(\)](#) and [format_battery\(\)](#) - the battery-grid equivalent of [make_table\(\)](#)'s one-call convenience, kept as a separate function rather than folded into [make_table\(\)](#) for the same dispatch-collision reason as [make_numeric_summary_table\(\)](#) (see this file's header note: `predictors = NULL` already routes [make_table\(\)](#) to a plain summary table, and a battery grid is also a no-predictors table).

Usage

```
make_battery_table(
  data,
  outcomes,
  statistics = "perc",
  conf = NULL,
  base = NULL,
  multicode = FALSE,
  formatted = TRUE
)
```

Arguments

<code>data</code>	A data frame, or survey design object.
<code>outcomes</code>	Character vector of categorical/multicoded outcome variable names sharing consistent category levels (e.g. a Likert battery).
<code>statistics</code>	Exactly one statistic code: "perc", "count", "w_perc", or "w_count".
<code>conf</code>	NULL, "se", or "ci" - inserted as an extra row under each item when requested.
<code>base</code>	Optional base/filter expression, passed to calc_stats() .
<code>multicode</code>	Passed to calc_stats() .
<code>formatted</code>	If TRUE (default), returns the styled huxtable. If FALSE, returns pivot_battery() 's own list output untouched (pivoted, stat_code, category_order, conf_type).

Value

A styled huxtable, or (if `formatted = FALSE`) the list returned by [pivot_battery\(\)](#).

See Also

[pivot_battery\(\)](#), [format_battery\(\)](#)

make_numeric_summary_table

Build a numeric-only summary table in one call

Description

Single entry point chaining [pivot_numeric_summary\(\)](#) straight to [format_numeric_summary\(\)](#) - the numeric-summary equivalent of [make_table\(\)](#)'s one-call convenience, kept as a separate function rather than folded into [make_table\(\)](#) (see this file's header note for the dispatch-collision reasoning: `predictors = NULL` already routes [make_table\(\)](#) to a plain summary table, and a numeric-statistics grid is also a no-predictors table).

Usage

```
make_numeric_summary_table(
  data,
  outcomes,
  statistics = c("mean", "sd"),
  weighted = FALSE,
  base = NULL,
  formatted = TRUE
)
```

Arguments

data	A data frame, or survey design object if <code>weighted = TRUE</code> .
outcomes	Character vector of numeric variable names to summarise.
statistics	One or more of mean, median, sum, min, max, range, iqr, sd. Default <code>c("mean", "sd")</code> .
weighted	If <code>TRUE</code> , requests the weighted equivalent of each statistic that has one (data must then be a survey design object).
base	Optional base/filter expression, passed to calc_stats() .
formatted	If <code>TRUE</code> (default), returns the styled huxtable. If <code>FALSE</code> , returns pivot_numeric_summary() 's own list output untouched.

Value

A styled huxtable, or (if `formatted = FALSE`) the list returned by [pivot_numeric_summary\(\)](#).

See Also

[pivot_numeric_summary\(\)](#), [format_numeric_summary\(\)](#)

make_table	<i>Single entry point: calc_stats() chained straight to the right pivot/format pair</i>
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Description

Dispatches on the shape of predictors to decide which table this is - a Total-only summary ([pivot_summary\(\)](#)), a flat crosstab ([pivot_crosstab\(\)](#)), or a single nested crosstab ([pivot_nested_crosstab\(\)](#)) - so a normal call site never has to name any of those three functions (or their `format_*()` partners) directly.

Usage

```
make_table(
  data,
  outcomes,
  predictors = NULL,
  statistics = c("count", "mean"),
  conf = NULL,
  base = NULL,
  pval = NULL,
  multicode = TRUE,
  pairwise = FALSE,
  formatted = TRUE
)
```

Arguments

data	A data frame, or survey design object.
outcomes	Character vector of outcome variable names.
predictors	NULL for a Total-only summary, a single variable name or list of variable names for a flat crosstab (one or more side-by-side predictor blocks), or <code>list(c(<outer>, <inner>))</code> for a single nested crosstab. Mixing a nested set with additional flat sets in one call is rejected - run two separate <code>make_table()</code> calls instead.
statistics	Character vector of statistic codes, passed to calc_stats() .
conf	NULL, "se", or "ci", passed to calc_stats() .
base	Optional base/filter expression, passed to calc_stats() .
pval	Optional p-value method, passed to calc_stats() .
multicode	Passed to calc_stats() .
pairwise	If TRUE, passed to calc_stats() - runs letter-based pairwise significance testing across predictor levels and adds the legend block/Sig. diff row wherever there's a predictor to compare across (pivot_crosstab() / pivot_nested_crosstab() and their <code>format_*()</code> partners already render this automatically). A no-op for a Total-only table (<code>predictors = NULL</code>), since there's nothing to compare.
formatted	If TRUE (default), returns the styled huxtable. If FALSE, stops after the pivot step and returns the matching <code>pivot_*</code> function's own list output untouched (no digit formatting or presentation-only rows), for further computation.

Value

A styled huxtable, or (if `formatted = FALSE`) the list returned by whichever `pivot_*`() function was dispatched to.

See Also

[calc_stats\(\)](#), [pivot_summary\(\)](#), [pivot_crosstab\(\)](#), [pivot_nested_crosstab\(\)](#)

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